Computer Networking

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Chapter 2
Application Layer

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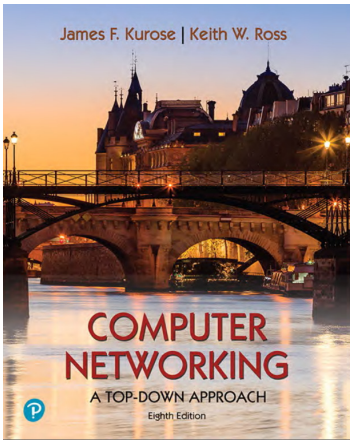


Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Assignments :

- Ch2 (ver8): 4,7,8,10,13,15,18,23,24,26,27
- A group project
- Key words:
  - client-server architecture, Transport Services, HTTP, Cookies, Non-Persistent and Persistent Connections, Caching and proxy, FTP, SMTP, POP3, IMAP, DNS, P2P Architectures

Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 FTP

2.4 electronic mail

- SMTP, POP3, IMAP

2.5 DNS

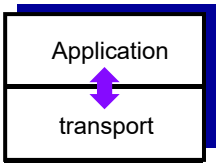
2.6 P2P applications

2.7 socket programming with UDP and TCP

Chapter 2: application layer

our goals:

- conceptual, implementation aspects of network application protocols
  - transport-layer service models
  - client-server paradigm
  - peer-to-peer paradigm



- learn about protocols by examining popular application-level protocols
  - HTTP
  - FTP
  - SMTP / POP3 / IMAP
  - DNS
- creating network applications
  - socket API

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Some network apps

- e-mail
- web
- text messaging
- remote login
- P2P file sharing
- multi-user network games
- streaming stored video (YouTube, Hulu, Netflix)
- voice over IP (e.g., Skype)
- real-time video conferencing
- social networking
- search
- …

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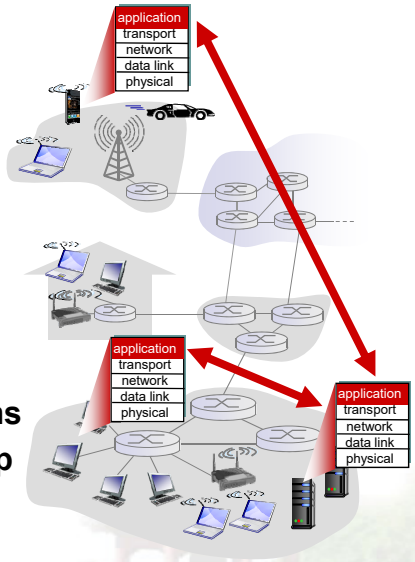
Creating a network app

write programs that:

- run on (different) *end systems*
- communicate over network
- e.g., web server software communicates with browser software

no need to write software for network-core devices

- network-core devices do not run user applications
- applications on end systems allows for rapid app development, propagation



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Application architectures

possible structure of applications:

- client-server (C/S)
- peer-to-peer (P2P)

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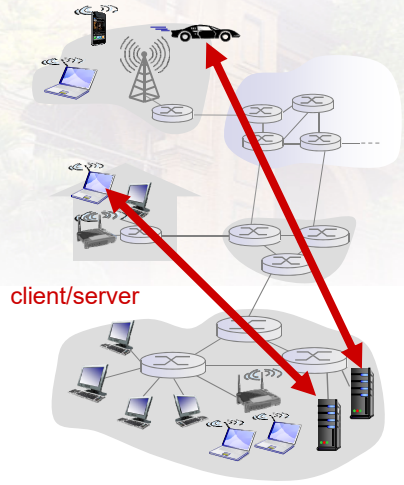
Client-server architecture



server:

- always-on host
- permanent IP address
- data centers for scaling

clients:

- communicate with server
- may be intermittently connected
- may have dynamic IP addresses
- do not communicate directly with each other

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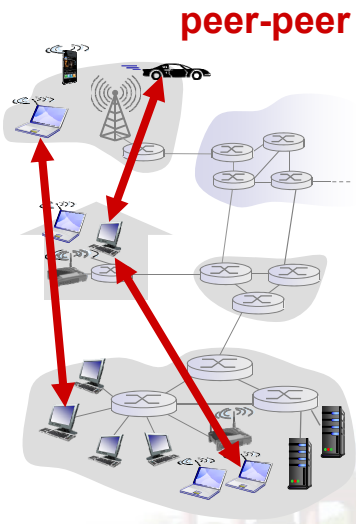
P2P architecture

- *no* always-on server
- arbitrary end systems directly communicate
- peers request service from other peers, provide service in return to other peers
  - ***self scalability*** **– new peers bring new service capacity, as well as new service demands**
- peers are intermittently connected and change IP addresses
  - complex management



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Processes communicating

process: **program running within a host**

- **within same host, two processes communicate using inter-process communication (defined by OS)**
- **processes in different hosts communicate by exchanging messages**

clients, servers

client process:
process that *initiates* communication

server process:
process that *waits to* be contacted

- **aside: applications with P2P architectures have client processes & server processes**

Communication: Process-To-Process

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Addressing processes

- **to receive messages, process must have *identifier***
- **host device has unique 32-bit IP address**
- ***Q:* does IP address of host on which process runs suffice for identifying the process?**
  - *A:* no, *many* processes can be running on same host
- **identifier includes both IP address and Port numbers associated with process on host.**

- **example port numbers:**
  - **HTTP server: 80**
  - **mail server: 25**
- **to send HTTP message to gaia.cs.umass.edu web server:**
  - **IP address: 128.119.245.12**
  - **port number: 80**

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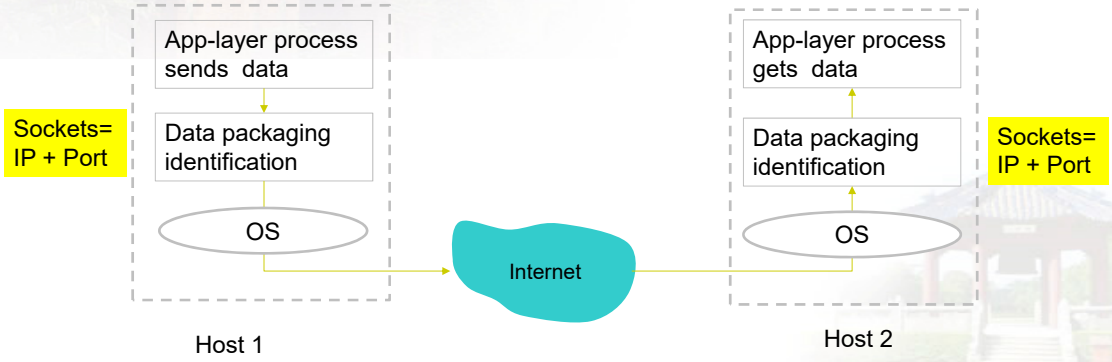
Sockets What is Socket?

- process sends/receives messages to/from its socket
- socket analogous to door
  - sending process shoves message out door
  - sending process relies on transport infrastructure on other side of door to deliver message to socket at receiving process



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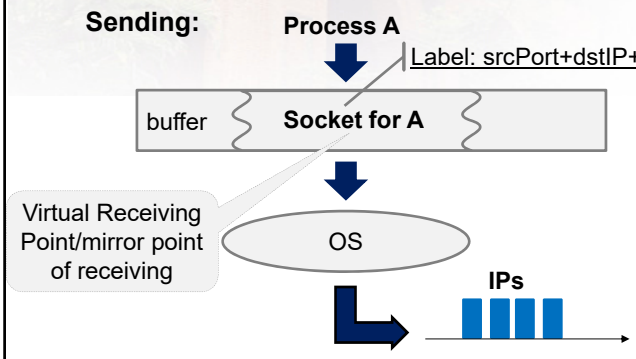
Sockets What is Socket?

- The essence of a socket is a special buffer for a process to send/receive data;
- The definition of a socket is related to its function



Questions:
- **What happens when a process sends data to multiple destination processes on different hosts at the same time?**
- **What happens when multiple processes on the same host send data to the same program on the same host at the same time?**

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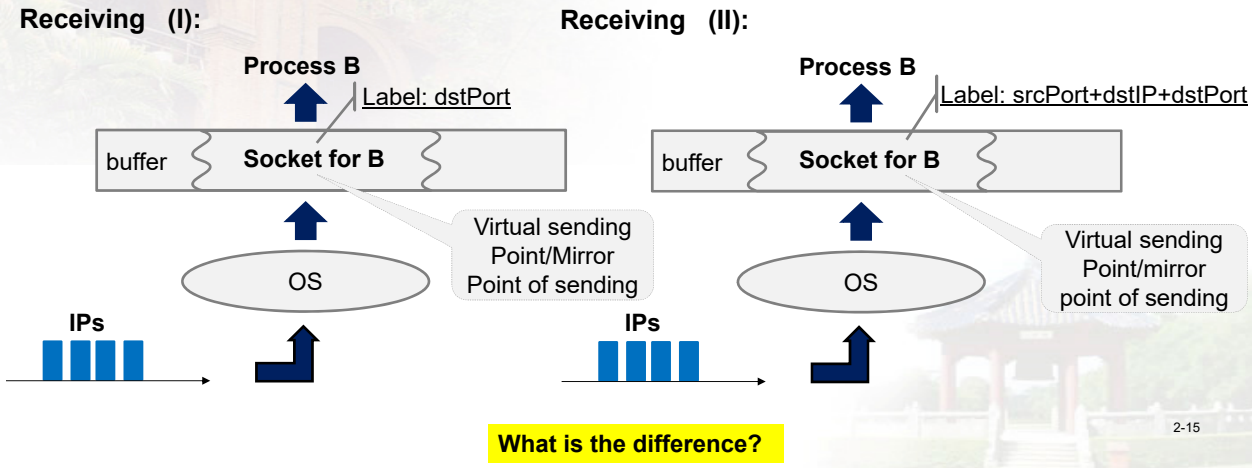
Sockets What is Socket?

- The essence of a socket is a special buffer for a process to send/receive data;
- The definition of a socket is related to its function



What is the difference?

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App-layer protocol defines

- **types of messages exchanged,**
  - **e.g., request, response**
- **message syntax:**
  - **what fields in messages & how fields are delineated**
- **message semantics**
  - **meaning of information in fields**
- **rules for when and how processes send & respond to messages**

open protocols:

- **defined in RFCs**
- **allows for interoperability**
- **e.g., HTTP, SMTP**

proprietary protocols:

- **e.g., Skype, QQ**

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What transport service does an app need?

data integrity

- **some apps (e.g., file transfer, web transactions) require 100% reliable data transfer**
- **other apps (e.g., audio) can tolerate some loss**

timing

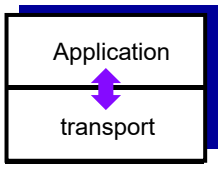
- **some apps (e.g., Internet telephony, interactive games) require low delay to be "effective"**

throughput

- some apps (e.g., multimedia) require minimum amount of throughput to be "effective"
- other apps ("elastic apps") make use of whatever throughput they get

security

- encryption, data integrity, …



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Transport service requirements: common apps

application	data loss	throughput	time sensitive
file transfer	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5kbps-1Mbps video:10kbps-5Mbps	yes, 100's msec
stored audio/video	loss-tolerant	same as above	yes, few secs
interactive games	loss-tolerant	few kbps up	yes, 100's msec
text messaging	no loss	elastic	yes and no

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Internet transport protocols services

TCP service:

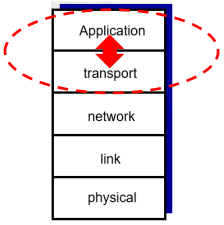
- *reliable transport* between sending and receiving process
- *flow control:* sender won't overwhelm receiver
- *congestion control:* throttle sender when network overloaded
- *connection-oriented:* setup required between client and server processes

- *DOES NOT provide:* timing, minimum throughput guarantee, security

UDP service:

- *unreliable data transfer* between sending and receiving process
- *does not provide:* reliability, flow control, congestion control, timing, throughput guarantee, security, or connection setup,

Q: why bother? Why is there a UDP?



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Internet apps: application, transport protocols

application	application layer protocol	underlying transport protocol
e-mail	SMTP [RFC 2821]	TCP
remote terminal access	Telnet [RFC 854]	TCP
Web	HTTP [RFC 2616]	TCP
file transfer	FTP [RFC 959]	TCP
streaming multimedia	HTTP (e.g., YouTube), RTP [RFC 1889]	TCP or UDP
Internet telephony	SIP, RTP, proprietary (e.g., Skype)	TCP or UDP

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Securing TCP

TCP & UDP

- no encryption
- Clear-text passwords sent into socket traverse Internet in clear-text

SSL

- provides encrypted TCP connection
- data integrity
- end-point authentication

SSL is at app layer

- Apps use SSL libraries, which "talk" to TCP

SSL socket API

- Clear-text passwords sent into socket traverse Internet encrypted
- See Chapter 7

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Chapter 2: outline

2.1 principles of network applications
- app architectures
- app requirements

2.2 Web and HTTP

2.3 FTP

2.4 electronic mail
- SMTP, POP3, IMAP

2.5 DNS

2.6 P2P applications

2.7 socket programming with UDP and TCP

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Web and HTTP

First, a review…

- ***web page*** **consists of** ***objects***
- **object can be HTML file, JPEG image, Java applet, audio file,…**
- **web page consists of** ***base HTML-file*** **which includes** ***several referenced objects***
- **each object is addressable by a** ***URL,***
  e.g., `www.someschool.edu/someDept/pic.gif`

host name — path name

HTTP overview http/1.1 (rfc2616)

HTTP: hypertext transfer protocol

- **Web's application layer protocol**
- **client/server model**
  - ***client:*** **browser that requests, receives, (using HTTP protocol) and "displays" Web objects**
  - ***server:*** **Web server sends (using HTTP protocol) objects in response to requests**

HTTP overview (continued)

uses TCP:

- client initiates TCP connection (creates socket) to server, port 80
- server accepts TCP connection from client
- HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP is "stateless"

- server maintains no information about past client requests

aside

protocols that maintain "state" are complex!

- past history (state) must be maintained
- if server/client crashes, their views of "state" may be inconsistent, must be reconciled

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HTTP connections

non-persistent HTTP

- at most one object sent over one TCP connection
  - connection then closed
- downloading multiple objects required multiple connections

persistent HTTP

- multiple objects can be sent over single TCP connection between client, server

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Non-persistent HTTP

suppose user enters URL:

`www.someSchool.edu/someDepartment/home.index`

(contains text, references to 10 jpeg images)

1a. HTTP client initiates TCP connection to HTTP server (process) at www.someSchool.edu on port 80

1b. HTTP server at host www.someSchool.edu waiting for TCP connection at port 80. "accepts" connection, notifying client

2. HTTP client sends HTTP *request message* (containing URL) into TCP connection socket. Message indicates that client wants object someDepartment/home.index

3. HTTP server receives request message, forms *response message* containing requested object, and sends message into its socket

time

2-30

Non-persistent HTTP (cont.)

4. HTTP server closes TCP connection.

5. HTTP client receives response message containing html file, displays html. Parsing html file, finds 10 referenced jpeg objects

6. Steps 1-5 repeated for **each** of 10 jpeg objects

time

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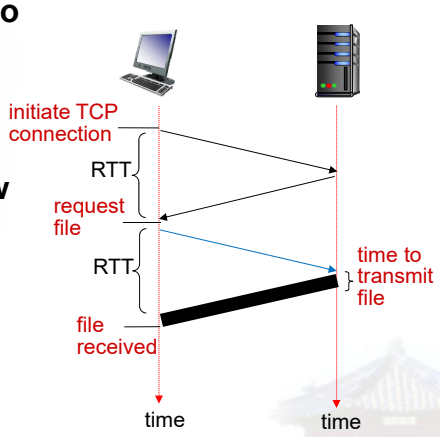
Non-persistent HTTP: response time

RTT (definition): time for a small packet to travel from client to server and back

HTTP response time:

- one RTT to initiate TCP connection
- one RTT for HTTP request and first few bytes of HTTP response to return
- file transmission time
- non-persistent HTTP response time =

2RTT+ file transmission time



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Persistent HTTP

non-persistent HTTP issues:

- requires 2 RTTs per object
- OS overhead for *each* TCP connection
- browsers often open parallel TCP connections to fetch referenced objects

persistent HTTP:

- server leaves connection open after sending response
- subsequent HTTP messages between same client/server sent over open connection
- client sends requests as soon as it encounters a referenced object
- as little as one RTT for all the referenced objects

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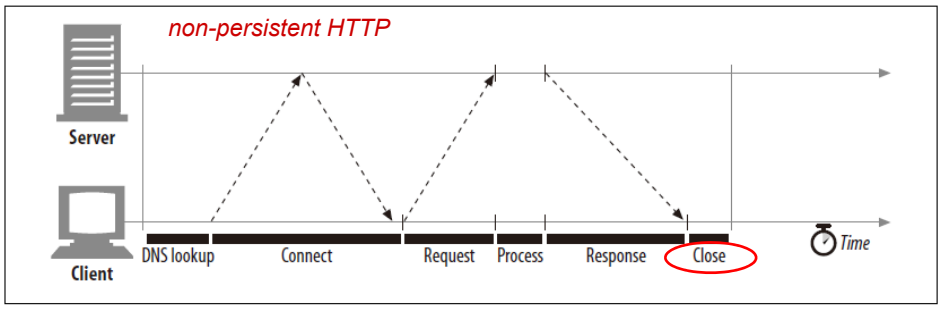

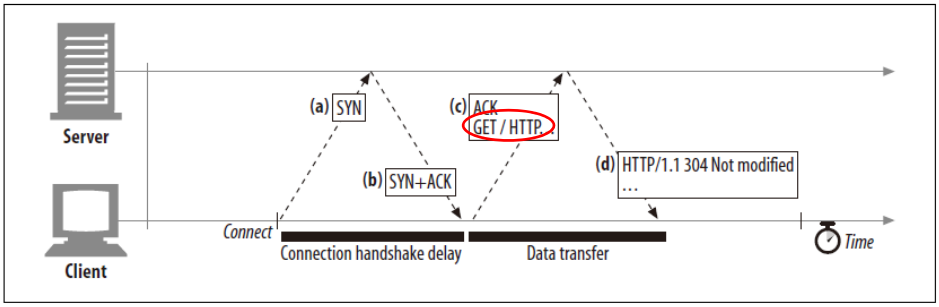

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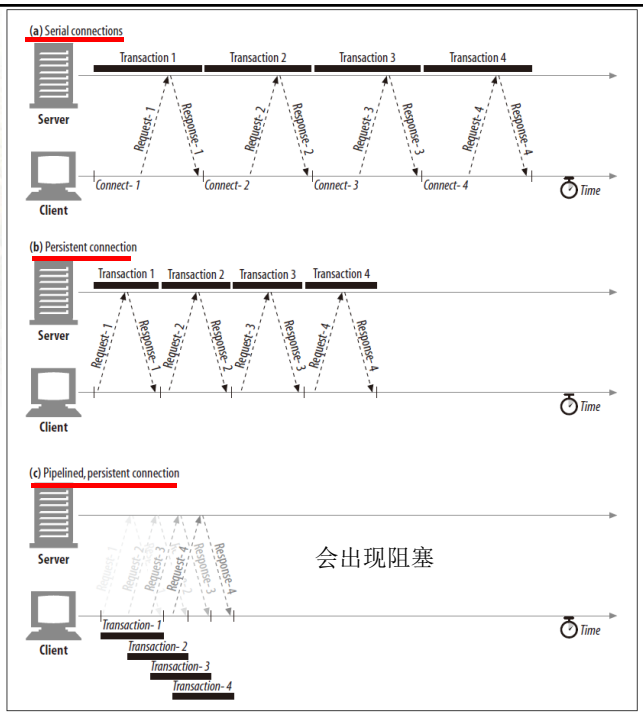

会出现阻塞

因为TCP导致排头阻塞，
HTTP3 采用UDP

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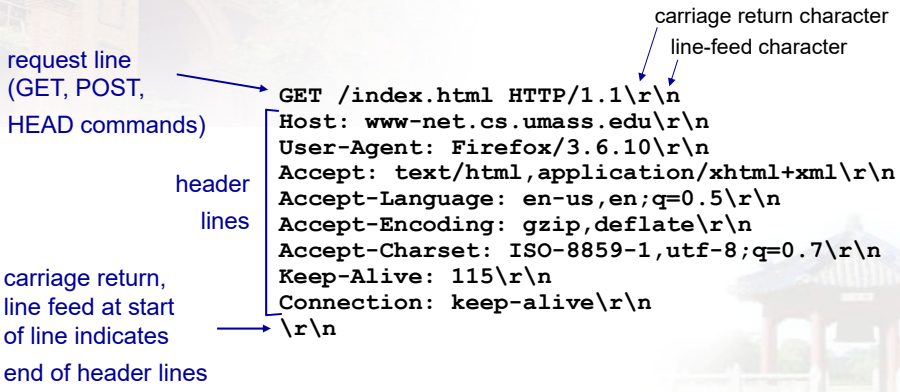
HTTP request message

- two types of HTTP messages: *request, response*
- HTTP request message:
  - ASCII (human-readable format)



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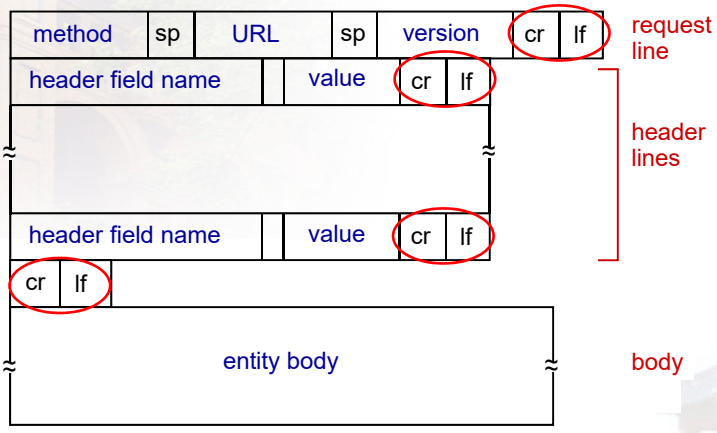
HTTP request message: general format



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Uploading form input

POST method:

- web page often includes form input
- input is uploaded to server in entity body
- Create new items

URL method:

- uses GET method
- input is uploaded in URL field of request line: `www.somesite.com/animalsearch?monkeys&banana`

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Method types

HTTP/1.0:

- GET
- POST
- HEAD
  - asks server to check requested object without responding it.

HTTP/1.1:

- GET, POST, HEAD
- PUT
  - uploads file in entity body to path specified in URL field
  - Modify (idempotent)
- DELETE
  - deletes file specified in the URL field

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HTTP response message

status line (protocol status code status phrase)

header lines

data, e.g., requested HTML file

```
HTTP/1.1 200 OK\r\n
Date: Sun, 26 Sep 2010 20:09:20 GMT\r\n
Server: Apache/2.0.52 (CentOS)\r\n
Last-Modified: Tue, 30 Oct 2007 17:00:02
   GMT\r\n
ETag: "17dc6-a5c-bf716880"\r\n
Accept-Ranges: bytes\r\n
Content-Length: 2652\r\n
Keep-Alive: timeout=10, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=ISO-8859-
   1\r\n
\r\n
data data data data data ...
```

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HTTP response status codes

- status code appears in 1st line in server-to-client response message.
- some sample codes:

200 OK
- request succeeded, requested object later in this msg

301 Moved Permanently
- requested object moved, new location specified later in this msg (Location:)

400 Bad Request
- request msg not understood by server

404 Not Found
- requested document not found on this server

505 HTTP Version Not Supported

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Trying out HTTP (client side) for yourself

1. Telnet to your favorite Web server:

```
telnet cis.poly.edu 80
```

opens TCP connection to port 80 (default HTTP server port) at cis.poly.edu. anything typed in sent to port 80 at cis.poly.edu

2. type in a GET HTTP request:

```
GET /~ross/ HTTP/1.1
Host: cis.poly.edu
```

by typing this in (hit carriage return twice), you send this minimal (but complete) GET request to HTTP server

3. look at response message sent by HTTP server!

(or use Wireshark to look at captured HTTP request/response)

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User-server state: cookies

many Web sites use cookies

four components:

1) **cookie header line of HTTP *response* message**
2) **cookie header line in next HTTP *request* message**
3) **cookie file kept on user's host, managed by user's browser**
4) **back-end database at Web site**

example:

- **Susan always access Internet from PC**
- **visits specific e-commerce site for first time**
- **when initial HTTP requests arrives at site, site creates:**
  - **unique ID**
  - **entry in backend database for ID**

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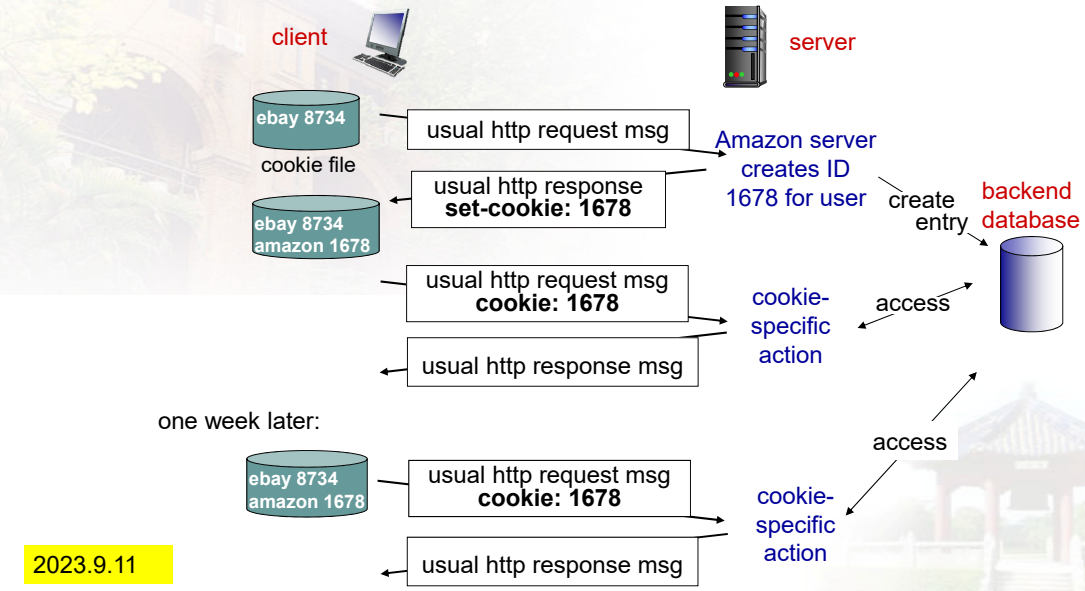
Cookies: keeping "state" (cont.)



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Cookies (continued)

what cookies can be used for:

- **authorization**
- **shopping carts**
- **recommendations**
- **user session state (Web e-mail)**

aside

cookies and privacy:

- cookies permit sites to learn a lot about you
- you may supply name and e-mail to sites

how to keep "state":

- protocol endpoints: maintain state at sender/receiver over multiple transactions
- cookies: http messages carry state

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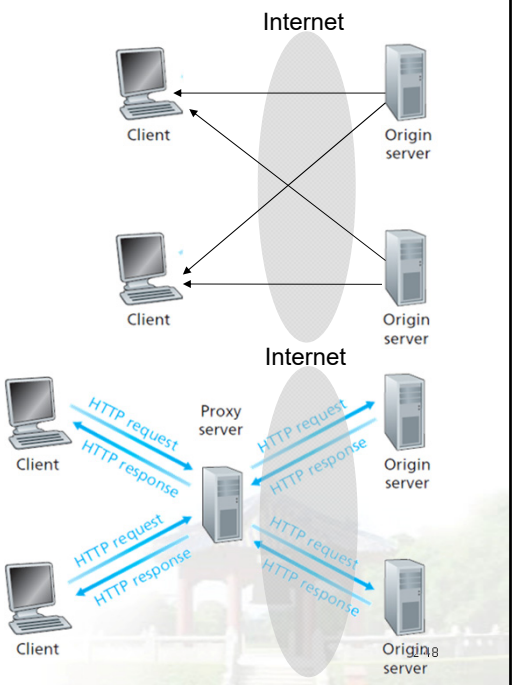
Web caches (proxy server)

goal: satisfy client request without involving origin server (就近访问)

- user sets browser: Web accesses via cache
- browser sends all HTTP requests to cache
  - object in cache: cache returns object
  - else cache requests object from origin server, then returns object to client



More about Web caching

- cache acts as both client and server
  - server for original requesting client
  - client to origin server
- typically cache is installed by ISP (university, company, residential ISP)

why Web caching?

- reduce response time for client request
- reduce traffic on an institution's access link
- Internet dense with caches: enables "poor" content providers to effectively deliver content (so too does P2P file sharing)

Caching example:

assumptions:

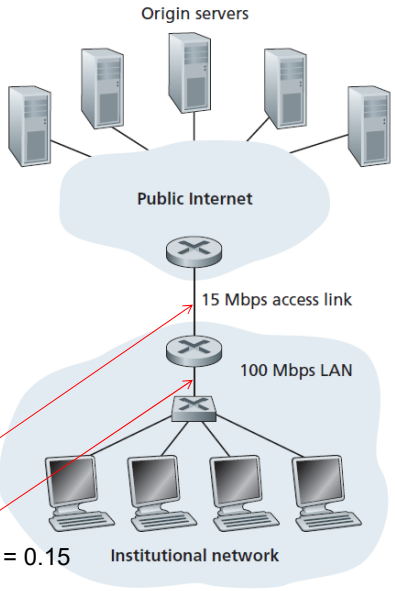
- avg object size: 1M bits
- avg request rate from browsers to origin servers:15/sec
- avg data rate to browsers: 15 Mbps
- RTT from institutional router to any origin server: 2 sec
- access link rate: 15 Mbps

consequences:

- LAN utilization: 15%
- access link utilization = 99% *problem!*
- total delay = Internet delay + access delay + LAN delay
  = 2 **sec** + **minutes** + **usecs**

(15 requests/sec) (1 Mbits/request)/(15 Mbps) = 1

(15 requests/sec) (1 Mbits/request)/(100 Mbps) = 0.15



瓶颈在哪里？

2-50

Caching example: fatter access link

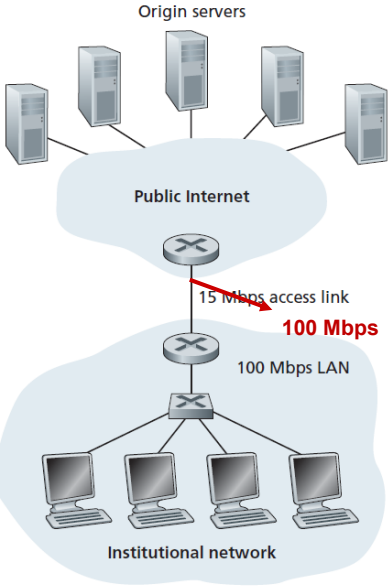
assumptions:

- avg object size: 1M bits
- avg request rate from browsers to origin servers:15/sec
- avg data rate to browsers: 15 Mbps
- RTT from institutional router to any origin server: 2 sec
- access link rate: ~~15 Mbps~~ → **100Mbps**

consequences:

- LAN utilization: 15%
- access link utilization = ~~99%~~ → **15%**
- total delay = Internet delay + access delay + LAN delay
  = 2 sec + minutes + usecs → **msecs**

Cost: increased access link speed (**not cheap**!)



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Caching example: install local cache

assumptions:

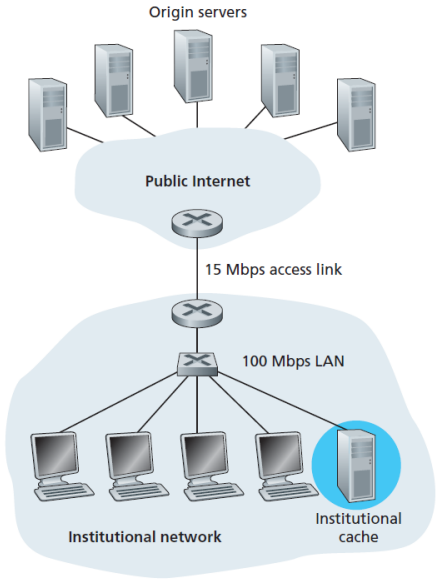
- avg object size: 1M bits
- avg request rate from browsers to origin servers:15/sec
- avg data rate to browsers: 15 Mbps
- RTT from institutional router to any origin server: 2 sec
- access link rate: 15 Mbps

consequences:

- LAN utilization: 15%
- access link utilization = ?
- total delay = Internet delay + access delay + LAN delay
  = 2 sec + minutes + usecs ?
  ?

Cost: web cache (cheap!)

How to compute link utilization, delay?



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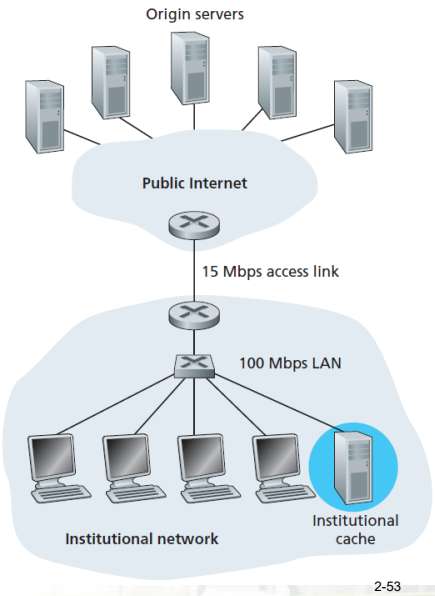
Caching example: install local cache

Calculating access link utilization, delay with cache:

- **suppose cache hit rate is 0.4**
  - **40% requests satisfied at cache, 60% requests satisfied at origin**
  - access link utilization:
    - 60% of requests use access link
  - data rate to browsers over access link = 0.6*15 Mbps = 9 Mbps
    - utilization = 9/15 = 0.6
  - total delay
    - **= 0.6 * (delay from origin servers) +0.4 * (delay when satisfied at cache)**
    - = 0.6 (2+0.01) + 0.4 (~msecs~0.01s)
    - = ~ 1.2 secs
    - less than with 100Mbps link (and cheaper too!)



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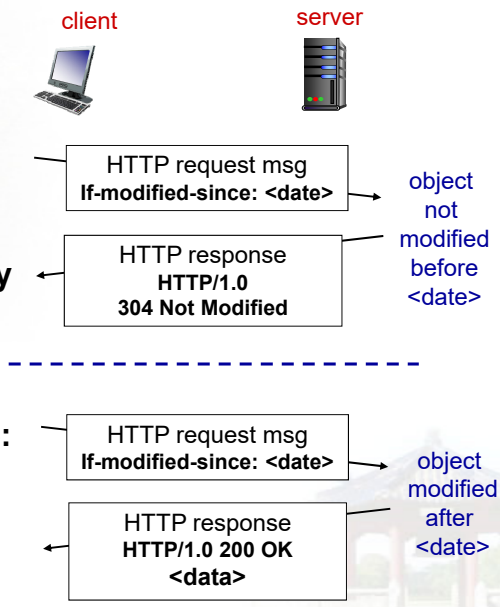
Conditional GET

- ***Goal:*** **don't send object if cache has up-to-date cached version**
  - **no object transmission delay**
  - **lower link utilization**
- ***cache:*** **specify date of cached copy in HTTP request**

  `If-modified-since: <date>`
- ***server:*** **response contains no object if cached copy is up-to-date:**

  `HTTP/1.0 304 Not Modified`



HTTP/2

Key goal: decreased delay in multi-object HTTP requests

HTTP1.1: introduced multiple, pipelined GETs over single TCP connection

- server responds *in-order* (FCFS: first-come-first-served scheduling) to GET requests
- with FCFS, small object may have to wait for transmission (head-of-line (HOL) blocking) behind large object(s)
- loss recovery (retransmitting lost TCP segments) stalls object transmission

HTTP/2

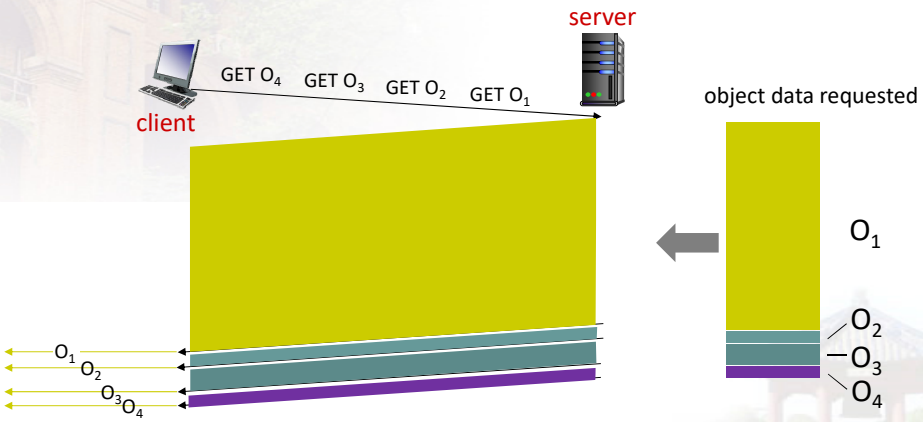
Key goal: decreased delay in multi-object HTTP requests

HTTP/2: [RFC 7540, 2015] increased flexibility at *server* in sending objects to client:

- methods, status codes, most header fields unchanged from HTTP 1.1
- transmission order of requested objects based on client-specified object priority (not necessarily FCFS)
- *push* unrequested objects to client
- divide objects into frames, schedule frames to mitigate HOL blocking

HTTP/2: mitigating HOL blocking

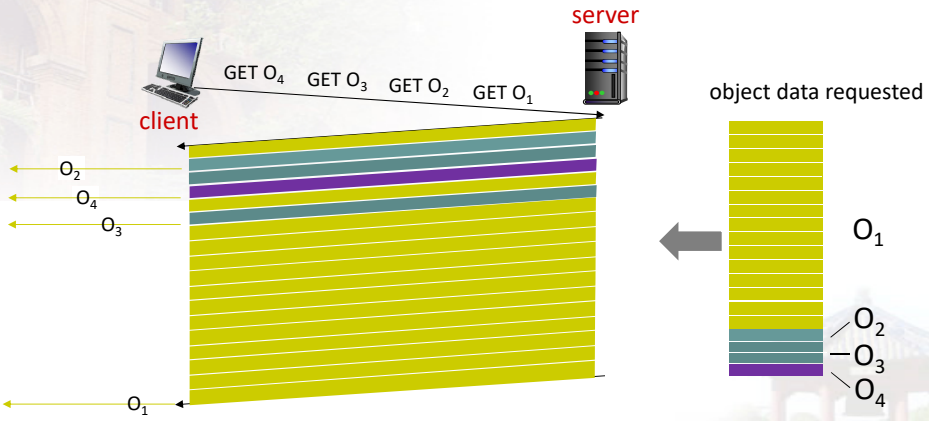
HTTP 1.1: client requests 1 large object (e.g., video file) and 3 smaller objects



objects delivered in order requested: O_2, O_3, O_4 wait behind O_1

HTTP/2: mitigating HOL blocking

HTTP/2: objects divided into frames, frame transmission interleaved



O_2, O_3, O_4 delivered quickly, O_1 slightly delayed

HTTP/2 to HTTP/3

HTTP/2 over single TCP connection means:

- recovery from packet loss still stalls all object transmissions
  - as in HTTP 1.1, browsers have incentive to open multiple parallel TCP connections to reduce stalling, increase overall throughput
- no security over vanilla TCP connection
- HTTP/3: adds security, per object error- and congestion-control (more pipelining) **over UDP**
  - more on HTTP/3 in transport layer

Chapter 2: outline

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- app architectures
- app requirements

2.2 Web and HTTP

2.3 FTP

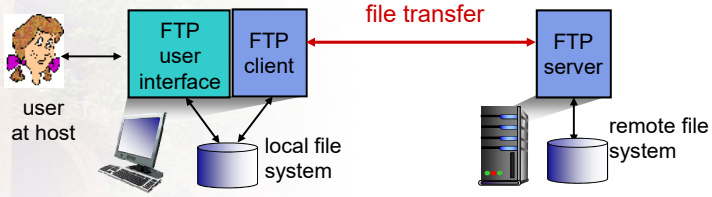
2.4 electronic mail
- SMTP, POP3, IMAP

2.5 DNS

2.6 P2P applications

2.7 socket programming with UDP and TCP

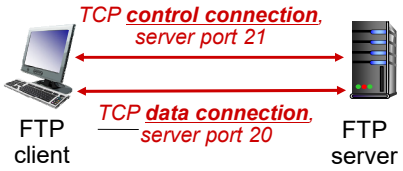
FTP: the file transfer protocol



- transfer file to/from remote host
- client/server model
  - *client:* side that initiates transfer (either to/from remote)
  - *server:* remote host
- ftp: RFC 959
- ftp server: port 21

FTP: separate control, data connections

- FTP client contacts FTP server at port 21, using TCP
- client authorized over control connection
- client browses remote directory, sends commands over control connection
- when server receives file transfer command, *server* opens *2nd* TCP data connection (for file) *to* client
- after transferring one file, server closes data connection



- server opens another TCP data connection to transfer another file
- control connection: *"out of band"*
- FTP server maintains "state": current directory, earlier authentication

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FTP commands, responses

sample commands:

- sent as ASCII text over control channel
- `USER` *username*
- `PASS` *password*
- `LIST` return list of file in current directory
- `RETR filename` retrieves (gets) file
- `STOR filename` stores (puts) file onto remote host

sample return codes

- status code and phrase (as in HTTP)
- `331 Username OK, password required`
- `125 data connection already open; transfer starting`
- `425 Can't open data connection`
- `452 Error writing file`

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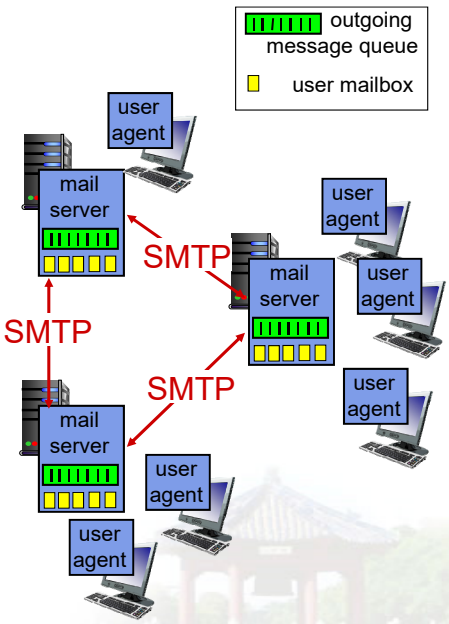
Electronic mail

Three major components:

- user agents
- mail servers
- simple mail transfer protocol: SMTP

User Agent

- a.k.a. "mail reader"
- composing, editing, reading mail messages
- e.g., Outlook, Thunderbird, iPhone mail client
- outgoing, incoming messages stored on server



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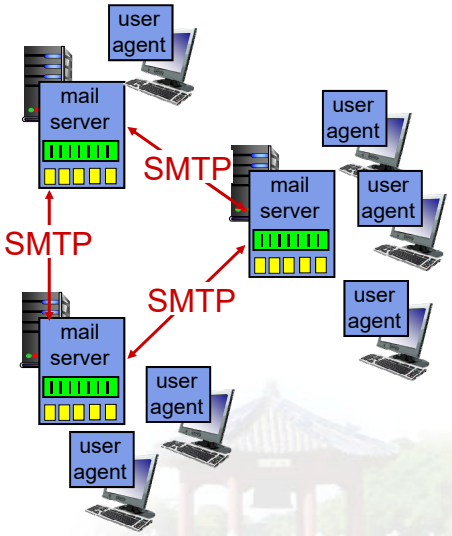
Electronic mail: mail servers

mail servers:

- ***mailbox*** **contains incoming messages for user**
- ***message queue*** **of outgoing (to be sent) mail messages**
- ***SMTP protocol*** **between mail servers to send email messages**
  - **client: sending mail server**
  - **"server": receiving mail server**



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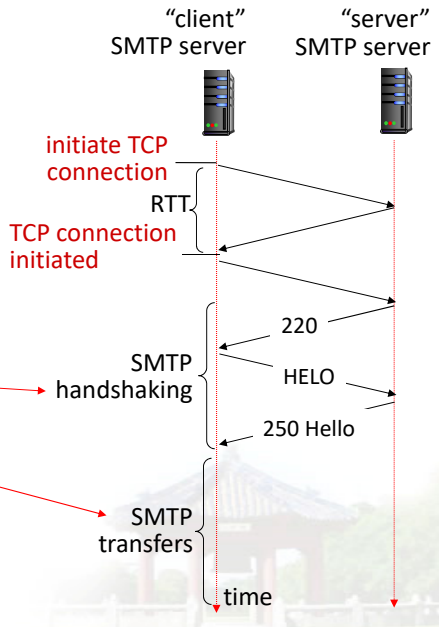
SMTP RFC (5321)

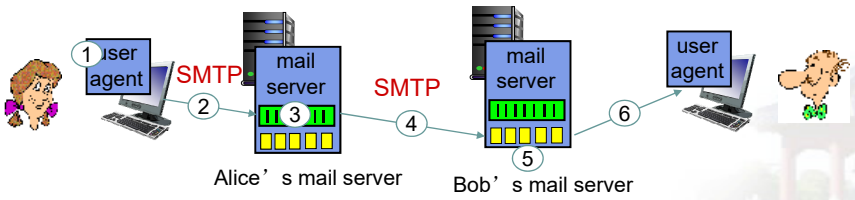
- uses TCP to reliably transfer email message from client (mail server initiating connection) to server, **port 25**
  - direct transfer: sending server (acting like client) to receiving server
- three phases of transfer
  - SMTP handshaking (greeting)
  - SMTP transfer of messages
  - SMTP closure
- command/response interaction (like HTTP)
  - commands: ASCII text
  - response: status code and phrase



Application Layer: 2-67

Scenario: Alice sends message to Bob

1) Alice uses UA to compose message "to" `bob@someschool.edu`
2) Alice's UA sends message to her mail server; message placed in message queue
3) client side of SMTP opens TCP connection with Bob's mail server
4) SMTP client sends Alice's message over the TCP connection
5) Bob's mail server places the message in Bob's mailbox
6) Bob invokes his user agent to read message



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Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

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Try SMTP interaction for yourself:

- `telnet servername 25`
- see 220 reply from server
- enter HELO, MAIL FROM, RCPT TO, DATA, QUIT commands

above lets you send email without using email client (reader)

SMTP: final words

- SMTP uses persistent connections
- SMTP requires message (header & body) to be in 7-bit ASCII
- SMTP server uses `CRLF.CRLF` to determine end of message

comparison with HTTP:

- HTTP: pull
- SMTP: push
- both have ASCII command/response interaction, status codes
- HTTP: each object encapsulated in its own response msg
- SMTP: multiple objects sent in multipart msg

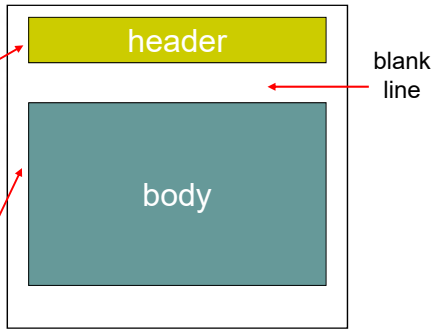
Mail message format

SMTP: protocol for exchanging email msgs

RFC 822: standard for text message format:

- header lines, e.g.,
  - To:
  - From:
  - Subject:

  different from SMTP MAIL FROM, RCPT TO: ***commands!***
- Body: the "message"
  - ASCII characters only



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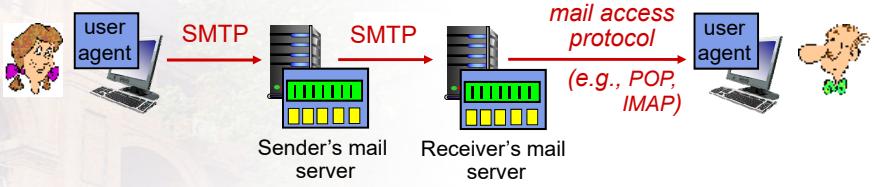
Mail access protocols



- SMTP: delivery/storage to receiver's server
- mail access protocol: retrieval from server
  - POP: Post Office Protocol [RFC 1939]: authorization, download
  - IMAP: Internet Mail Access Protocol [RFC 1730]: more features, including manipulation of stored msgs on server
  - HTTP: gmail, Hotmail, Yahoo! Mail, etc.

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POP3 protocol

authorization phase

- client commands:
  - `user`: declare username
  - `pass`: password
- server responses
  - `+OK`
  - `-ERR`

transaction phase, client:

- `list`: list message numbers, size
- `retr`: retrieve message by number
- `dele`: delete
- `quit`

```
S: +OK POP3 server ready
C: user bob
S: +OK
C: pass hungry
S: +OK user successfully logged on
```

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: <message 1 contents>
S: .
C: dele 1
C: retr 2
S: <message 1 contents>
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

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POP3 (more) and IMAP

more about POP3

- previous example uses POP3 "download and delete" mode
  - Bob cannot re-read e-mail if he changes client
- POP3 "download-and-keep": copies of messages on different clients
- POP3 is stateless across sessions

IMAP

- keeps all messages in one place: at server
- allows user to organize messages in folders
- keeps user state across sessions:
  - names of folders and mappings between message IDs and folder name

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DNS: domain name system

people: many identifiers:
- SSN, name, passport #

Internet hosts, routers:
- IP address (32 bit) - used for addressing datagrams
- "name", e.g., www.yahoo.com - used by humans

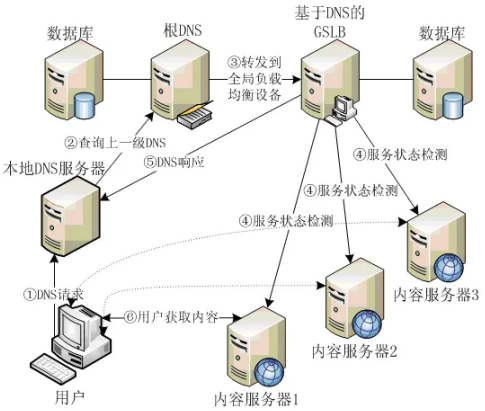
Q: how to map between IP address and name, and vice versa ?

Domain Name System:
- *distributed database* implemented in hierarchy of many *name servers*
- *application-layer protocol:* hosts, name servers communicate to *resolve* names (address/name translation)
  - note: core Internet function, implemented as application-layer protocol
  - complexity at network's "edge"

DNS: services, structure

DNS services

- **hostname to IP address translation**
- **host aliasing**
  - **canonical, alias names**
- **mail server aliasing**
- **load distribution**
  - **replicated Web servers: many IP addresses correspond to one name**



why not centralize DNS?

- **single point of failure**
- **traffic volume**
- **distant centralized database**
- **maintenance**

A: doesn't scale!

Thinking about the DNS

humongous distributed database:

- ~ billion records, each simple

handles many *trillions* of queries/day:

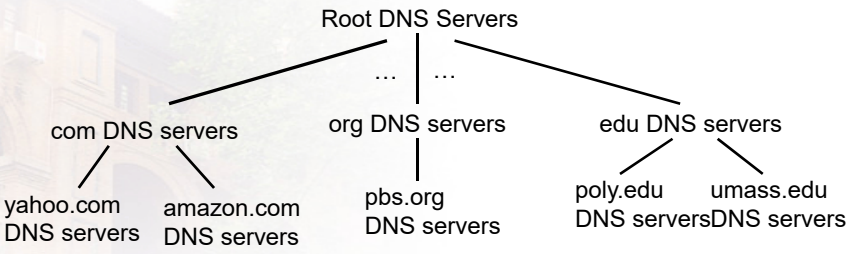
- *many* more reads than writes
- *performance matters:* almost every Internet transaction interacts with DNS - msecs count!

organizationally, physically decentralized:

- millions of different organizations responsible for their records

"bulletproof": reliability, security

DNS: a distributed, hierarchical database



client wants IP for www.amazon.com; 1st approx:

- **client queries root server to find com DNS server**
- **client queries .com DNS server to get amazon.com DNS server**
- **client queries amazon.com DNS server to get IP address for www.amazon.com**

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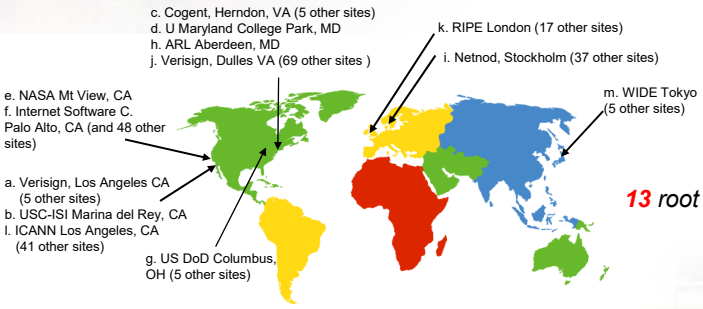
DNS: root name servers

- **contacted by local name server that can not resolve name**
- **root name server:**
  - **contacts authoritative name server if name mapping not known**
  - **gets mapping**
  - **returns mapping to local name server**



***13** root name "servers" worldwide*

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IPv4根服务器分布表

名称	地位	主服务器运营者[1]	主服务器位置[1]	IP[1]
A	DM and Root server	INTERNI.NET	美国弗吉尼亚州	198.41.0.4
B	Root server	美国信息科学研究所	美国加利弗尼亚州	128.9.0.107
C	Root server	PSINet公司	美国弗吉尼亚州	192.33.4.12
D	Root server	马里兰大学	美国马里兰州	128.8.10.90
E	Root server	美国航空航天管理局	美国加利弗尼亚州	192.203.230.10
F	Root server	因特网软件联盟	美国加利弗尼亚州	192.5.5.241
G	Root server	美国国防部网络信息中心	美国弗吉尼亚州	192.112.36.4
H	Root server	美国陆军研究所	美国马里兰州	128.63.2.53
I	Root server	Autonomica公司	瑞典斯德哥尔摩	192.36.148.17
J	Root server	VerSign公司	美国弗吉尼亚州	192.58.128.30
K	Root server	RIPE NCC	英国伦敦	192.0.14.129
L	Root server	IANA	美国弗吉尼亚州	198.32.64.12
M	Root server	WIDE Project	日本东京	202.12.27.33

全世界只有13台IPv4根域名服务器。1个为主根服务器在美国。其余12个均为辅根服务器，其中9台在美国，欧洲2个，位于英国和瑞典，亚洲1个位于日本。

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“雪人计划”IPv6根服务器全球分布情况

国家	主根服务器	辅根服务器	国家	主根服务器	辅根服务器
中国	1	3	西班牙	0	1
美国	1	2	奥地利	0	1
日本	1	0	智利	0	1
印度	0	3	南非	0	1
法国	0	3	澳大利亚	0	1
德国	0	2	瑞士	0	1
俄罗斯	0	1	荷兰	0	1
意大利	0	1			

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TLD, authoritative servers

top-level domain (TLD) servers: .com DNS server

- responsible for com, org, net, edu, aero, jobs, museums, and all top-level country domains, e.g.: uk, fr, ca, jp
- Network Solutions maintains servers for .com TLD
- Educause for .edu TLD

authoritative DNS servers: amazon.com DNS server

- Organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts
- can be maintained by organization or service provider

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Local DNS name server

- does not strictly belong to hierarchy
- each ISP (residential ISP, company, university) has one
  - also called "default name server"
- when host makes DNS query, query is sent to its local DNS server
  - has local cache of recent name-to-address translation pairs (but may be out of date!)
  - acts as proxy, forwards query into hierarchy

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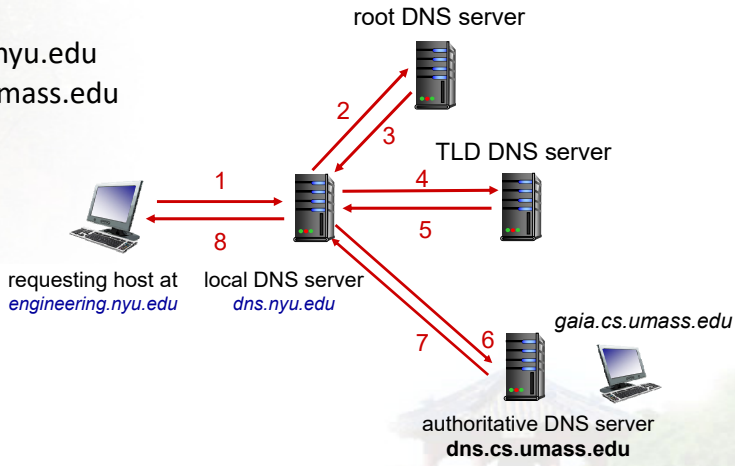
DNS name resolution: iterated query

Example: host at engineering.nyu.edu wants IP address for gaia.cs.umass.edu

Iterated query:

- contacted server replies with name of server to contact
- "I don't know this name, but ask this server"



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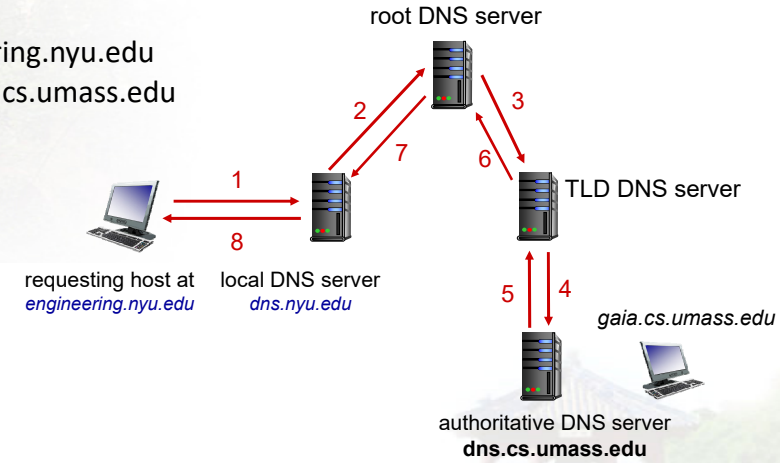
DNS name resolution: recursive query

Example: host at engineering.nyu.edu wants IP address for gaia.cs.umass.edu

Recursive query:

- puts burden of name resolution on contacted name server
- heavy load at upper levels of hierarchy?



DNS: caching, updating records

- once (any) name server learns mapping, it *caches* mapping
  - cache entries timeout (disappear) after some time (TTL)
  - TLD servers typically cached in local name servers
    - thus root name servers not often visited
- cached entries may be *out-of-date* (best effort name-to-address translation!)
  - if name host changes IP address, may not be known Internet-wide until all TTLs expire
- update/notify mechanisms proposed IETF standard
  - RFC 2136

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DNS records

DNS: **distributed db storing resource records (RR)**

RR format: (`name`, `value`, `type`, `ttl`)

type=A

- `name` is hostname
- `value` is IP Address

type=NS

- `name` is domain (e.g., foo.com)
- `value` is hostname of authoritative Name Server for this domain

type=CNAME

- `name` is alias name for some "canonical" (the real) name
- `e.g., www.ibm.com` is really `servereast.backup2.ibm.com`
- `value` is Canonical NAME

type=MX

- `value` is name of Mailserver associated with `name`

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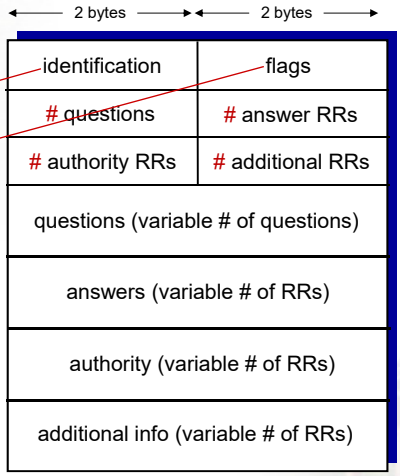
DNS protocol, messages

- ***query*** **and** ***reply*** **messages, both with same** ***message format***

msg header

- identification: 16 bit # for query, reply to query uses same #
- flags:
  - query or reply
  - recursion desired
  - recursion available
  - reply is authoritative



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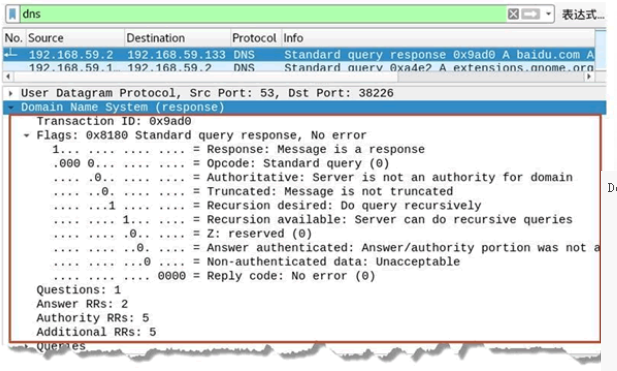
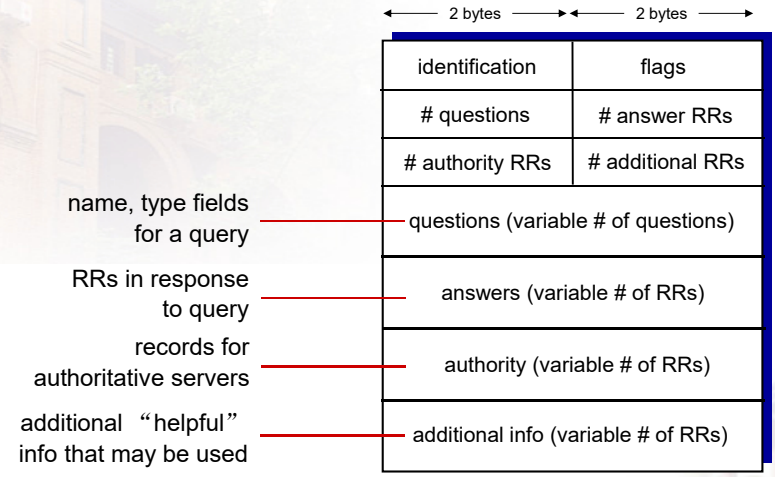



```
Domain Name System (response)
    Transaction ID: 0x9ad0                                  #事务ID
    Flags: 0x8180 Standard query response, No error         #报文中的标志字段
        1... .... .... .... = Response: Message is a response
                                                            #QR字段，值为1，因为是一个响应包
        .000 0... .... .... = Opcode: Standard query (0)    # Opcode字段
        .... .0.. .... .... = Authoritative: Server is not an authority for
        domain                                              #AA字段
        .... ..0. .... .... = Truncated: Message is not truncated
                                                            #TC字段
        .... ...1 .... .... = Recursion desired: Do query recursively
                                                            #RD字段
        .... .... 1... .... = Recursion available: Server can do recursive
        queries                                             #RA字段
        .... .... .0.. .... = Z: reserved (0)
        .... .... ..0. .... = Answer authenticated: Answer/authority portion
        was not authenticated by the server
        .... .... ...0 .... = Non-authenticated data: Unacceptable
        .... .... .... 0000 = Reply code: No error (0)      #返回码字段
    Questions: 1
    Answer RRs: 2
    Authority RRs: 5
    Additional RRs: 5
```

DNS protocol, messages



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Inserting records into DNS

- example: new startup "Network Utopia"
- register name networkuptopia.com at *DNS registrar* (e.g., Network Solutions)
  - provide names, IP addresses of authoritative name server (primary and secondary)
  - registrar inserts two RRs into .com TLD server:
    `(networkutopia.com, dns1.networkutopia.com, NS)`
    `(dns1.networkutopia.com, 212.212.212.1, A)`
- create authoritative server type A record for www.networkuptopia.com; type MX record for networkutopia.com

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Attacking DNS

DDoS attacks

- Bombard root servers with traffic
  - Not successful to date
  - Traffic Filtering
  - Local DNS servers cache IPs of TLD servers, allowing root server bypass
- Bombard TLD servers
  - Potentially more dangerous

Redirect attacks

- Man-in-middle
  - Intercept queries
- DNS poisoning
  - Send bogus relies to DNS server, which caches

Exploit DNS for DDoS

- Send queries with spoofed source address: target IP
- Requires amplification

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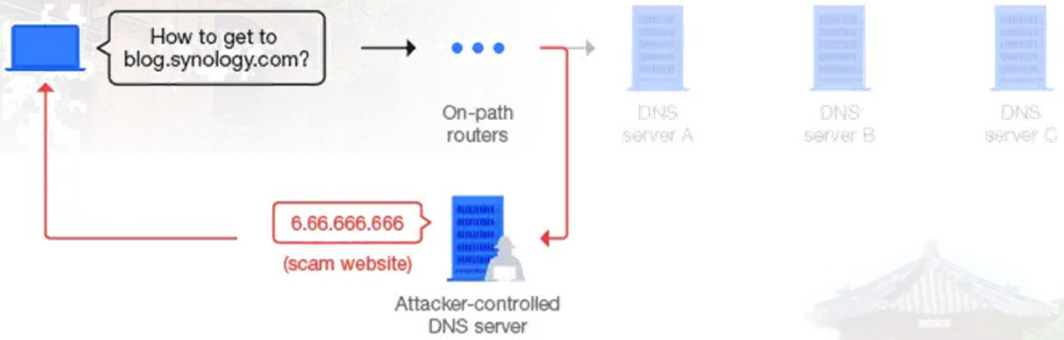
DNS诞生于1983年，在几次修订后最终于1987年定型。整个查询过程以TCP协议或UDP协议在53端口上进行，并且以明文进行查询。整个过程完全没有验证、没有隐私可言，这意味着任何人都可以对DNS查询过程进行攻击与篡改。



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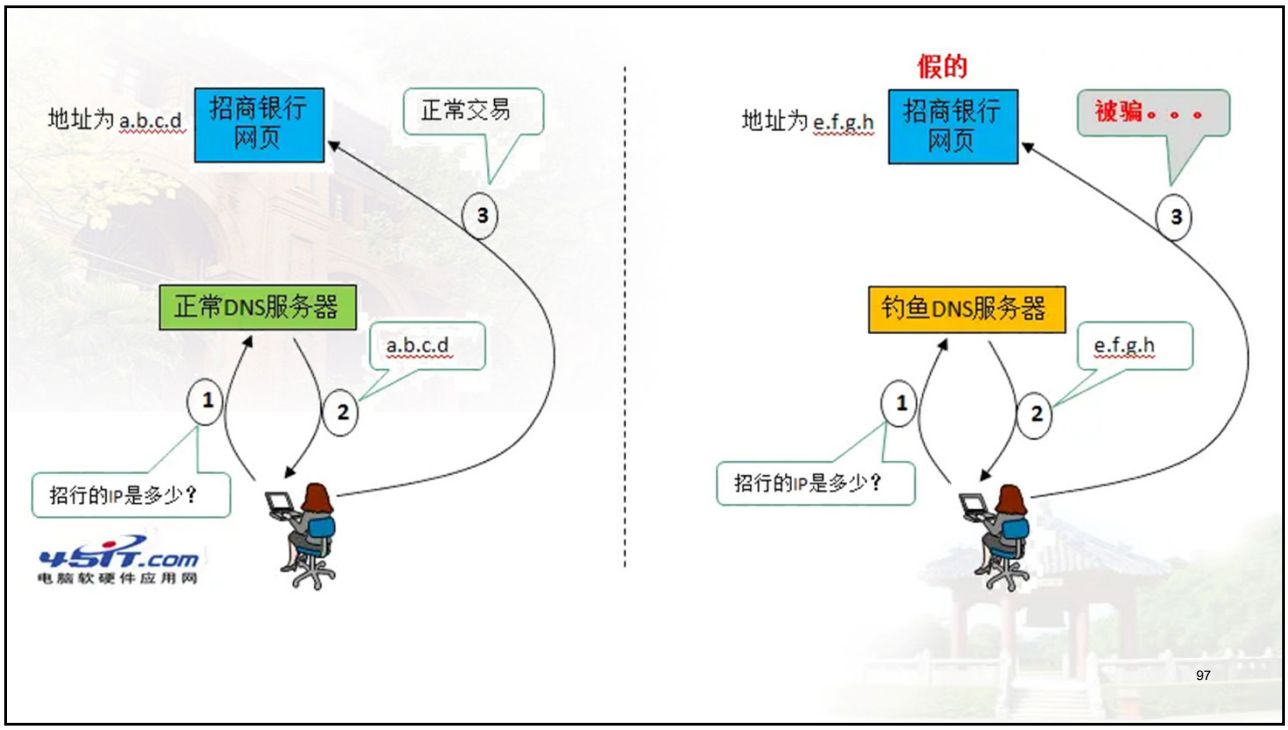
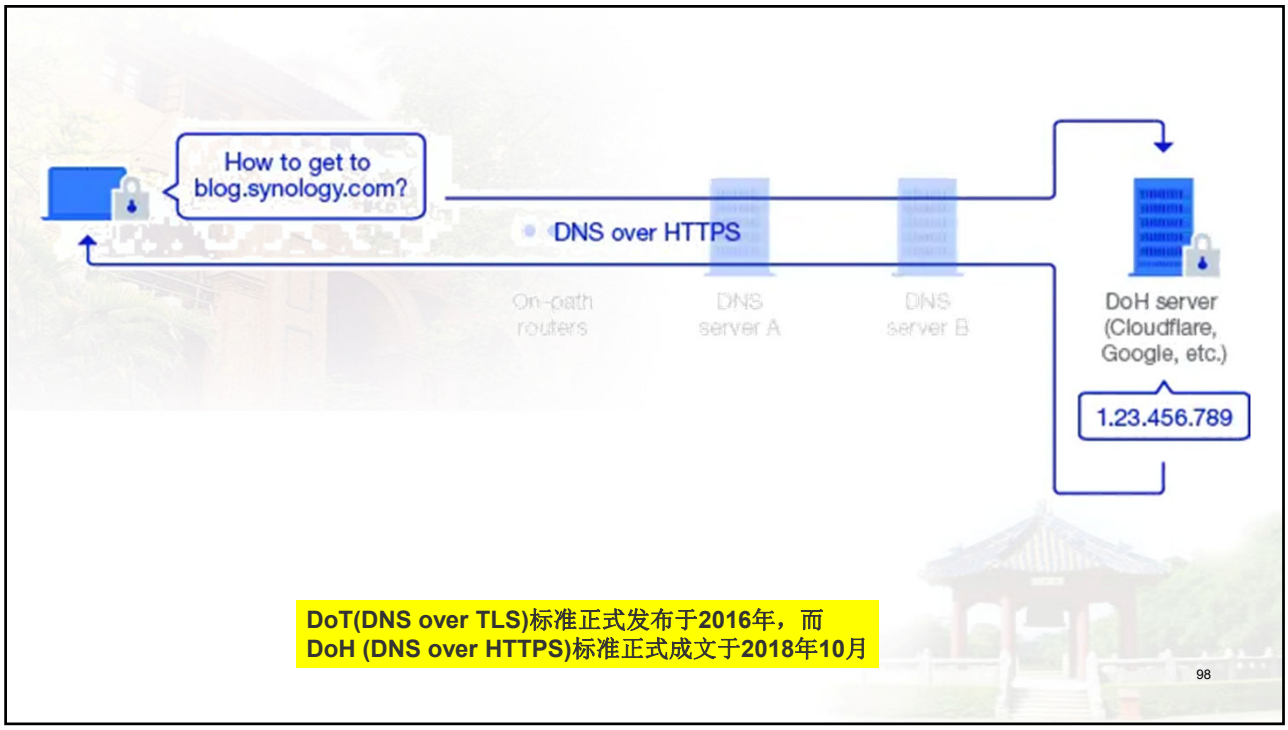




DoT(DNS over TLS)标准正式发布于2016年，而
DoH (DNS over HTTPS)标准正式成文于2018年10月

Chapter 2: outline

2.1 principles of network applications
- app architectures
- app requirements

2.2 Web and HTTP

2.3 FTP

2.4 electronic mail
- SMTP, POP3, IMAP

2.5 DNS

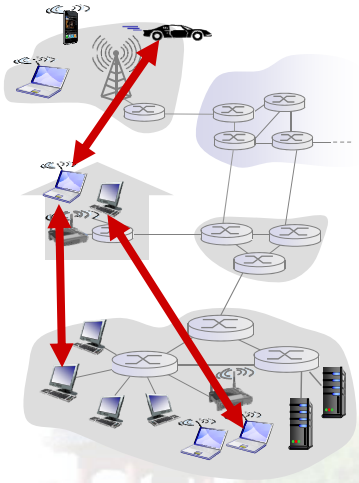
2.6 P2P applications

2.7 socket programming with UDP and TCP

Pure P2P architecture

- *no* always-on server
- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses

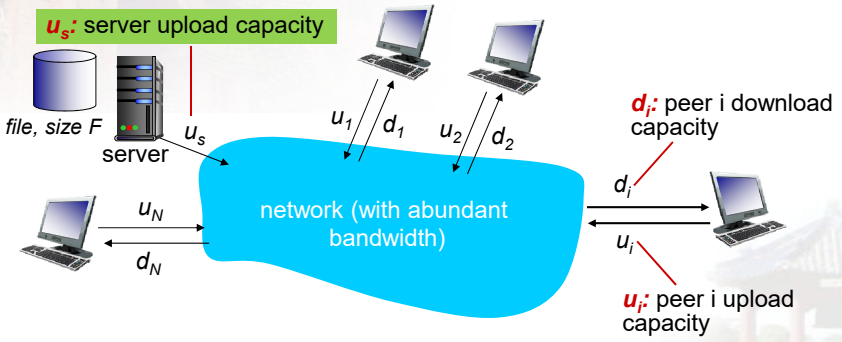
examples:
- file distribution (BitTorrent)
- Streaming (KanKan)
- VoIP (Skype)



File distribution: client-server vs P2P

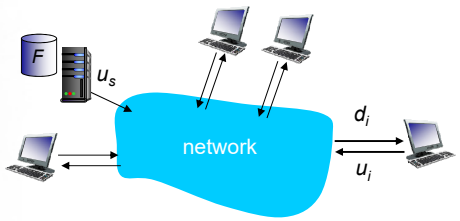
Question: **how much time to distribute file (size *F*) from one server to *N* peers?**

- **peer upload/download capacity is limited resource**



File distribution time: client-server

- ***server transmission:*** **must sequentially send (upload) *N* file copies:**
  - **time to send one copy: F/u_s**
  - **time to send N copies: NF/u_s**



  - *client:* each client must download file copy
    - d_{min} = min client download rate
    - min client download time: F/d_{min}

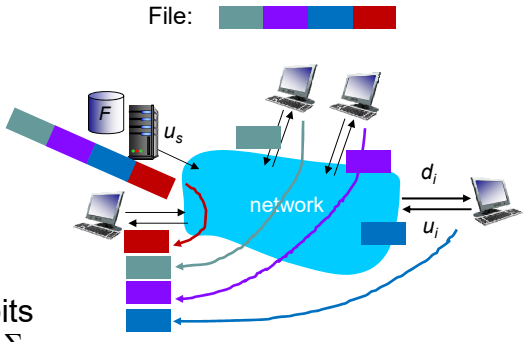
time to distribute F to N clients using client-server approach

$$D_{c\text{-}s} \geq max\{NF/u_{s,}, F/d_{min}\}$$

increases linearly in N

File distribution time: P2P

- ***server transmission:*** **must upload at least one copy**
  - **time to send one copy: F/u_s**
  - ***client:*** each client must download file copy
    - min client download time: F/d_{min}
  - ***clients:*** as aggregate must download *NF* bits
    - max upload rate (limiting max download rate) is $u_s + \Sigma u_i$



time to distribute F to N clients using P2P approach

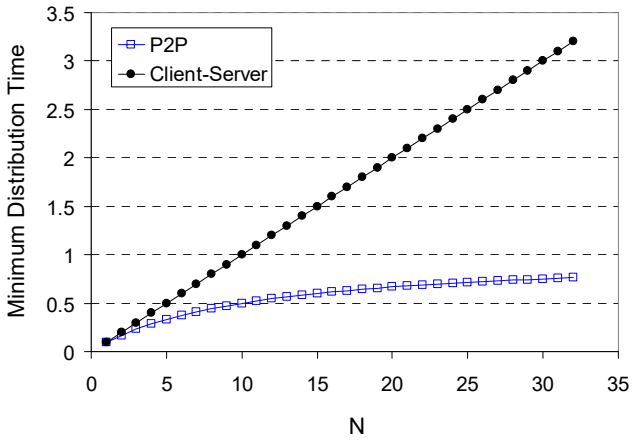
$$D_{P2P} \geq max\{F/u_s, F/d_{min}, NF/(u_s + \Sigma u_i)\}$$

increases linearly in *N* …

… but so does this, as each peer brings service capacity

Client-server vs. P2P: example

client upload rate = u, F/u = 1 hour, $u_s = 10u$, $d_{min} \geq u_s$

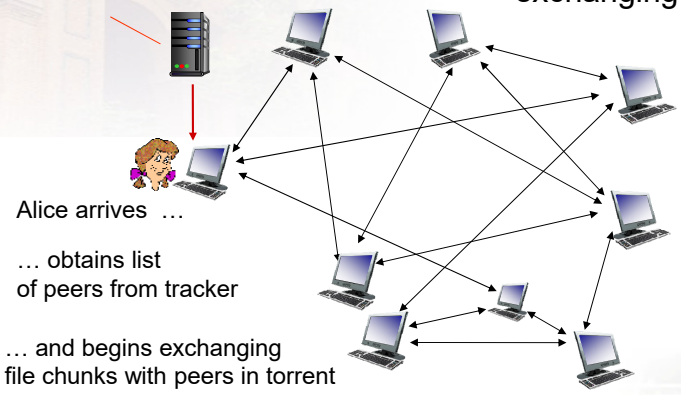


P2P file distribution: BitTorrent

- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



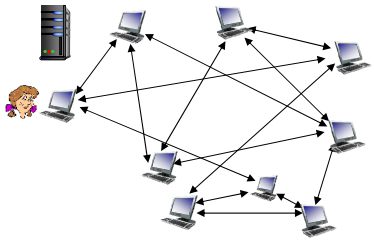
Alice arrives …

… obtains list of peers from tracker

… and begins exchanging file chunks with peers in torrent

P2P file distribution: BitTorrent

- **peer joining torrent:**
  - **has no chunks, but will accumulate them over time from other peers**
  - **registers with tracker to get list of peers, connects to subset of peers ("neighbors")**



- while downloading, peer uploads chunks to other peers
- peer may change peers with whom it exchanges chunks
- *churn:* peers may come and go
- once peer has entire file, it may (selfishly) leave or (altruistically) remain in torrent

BitTorrent: requesting, sending file chunks

requesting chunks:

- **at any given time, different peers have different subsets of file chunks**
- **periodically, Alice asks each peer for list of chunks that they have**
- **Alice requests missing chunks from peers, rarest first**

让稀缺资源变多

sending chunks: tit-for-tat

- Alice sends chunks to those four peers currently sending her chunks *at highest rate*
  - other peers are choked by Alice (do not receive chunks from her)
  - re-evaluate top 4 every10 secs
- every 30 secs: randomly select another peer, starts sending chunks
  - "optimistically unchoke" this peer
  - newly chosen peer may join top 4

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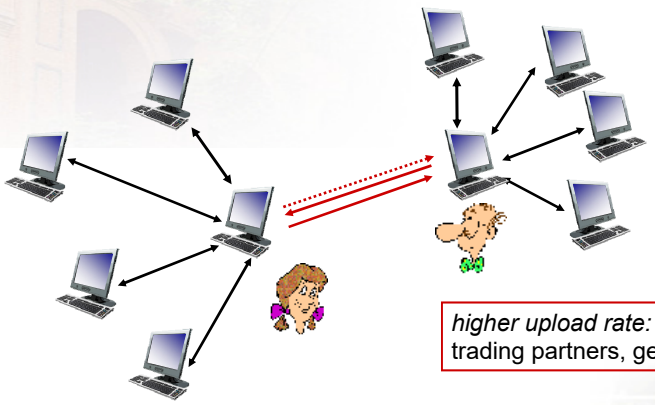
BitTorrent: tit-for-tat

(1) Alice "optimistically unchokes" Bob

(2) Alice becomes one of Bob's top-four providers; Bob reciprocates

(3) Bob becomes one of Alice's top-four providers



higher upload rate: find better trading partners, get file faster !

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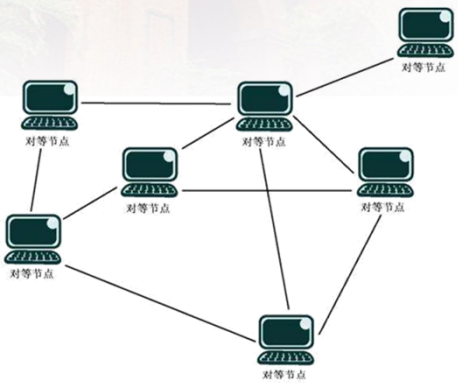
Distributed Hash Table (DHT)

- Hash table
- DHT paradigm
- Circular DHT and overlay networks
- Peer churn

Distributed Hash Table (DHT)

- Background
  - How to distribute the ***chunks*** to peers and manage the (peer, chunk)

Data Fragmentation



Simple Database

Simple database with(key, value) pairs:

- key: human name; value: social security #

Original Key	Value
John Washington	132-54-3570
Diana Louise Jones	761-55-3791
Xiaoming Liu	385-41-0902
Rakesh Gopal	441-89-1956
Linda Cohen	217-66-5609
.....	.....
Lisa Kobayashi	177-23-0199

key: movie title;
value: movie data.

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Hash Table

- More convenient to store and search on numerical representation of key
- key = hash(original key)

Original Key	Hash Key	Value
John Washington	8962458	132-54-3570
Diana Louise Jones	7800356	761-55-3791
Xiaoming Liu	1567109	385-41-0902
Rakesh Gopal	2360012	441-89-1956
Linda Cohen	5430938	217-66-5609
.....		.....
Lisa Kobayashi	9290124	177-23-0199

key: hash(movie title);
value: movie data.

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Distributed Hash Table (DHT)

- Distribute (key, value) pairs over millions of peers
  - pairs are evenly distributed over peers
- Any peer can query database with a key
  - database returns value for the key
  - To resolve query, small number of messages exchanged among peers
- Each peer only knows about a small number of other peers
- Robust to peers coming and going (churn)

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Assign key-value pairs to peers

- rule: assign (key, value) pair to the peer that has the *closest* ID.
- convention: closest is the *immediate successor* of the key.
  - e.g., ID space of peers is {0,1,2,3,…,63}
    suppose 8 peers whose IDs are: 1,12,13,25,32,40,48,60
  - If key = 51, then assigned to peer 60
  - If key = 60, then assigned to peer 60
  - If key = 61, then assigned to peer 1

HASH the data's name

Peer ID

Data ID vs. peer ID ?

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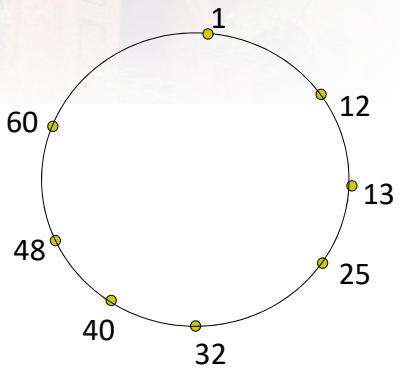
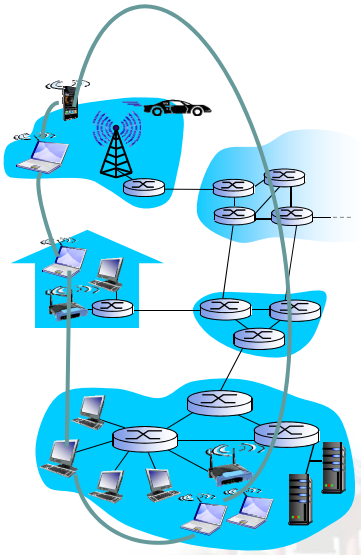
Circular DHT

- each peer *only* aware of immediate successor and predecessor.





"overlay network"

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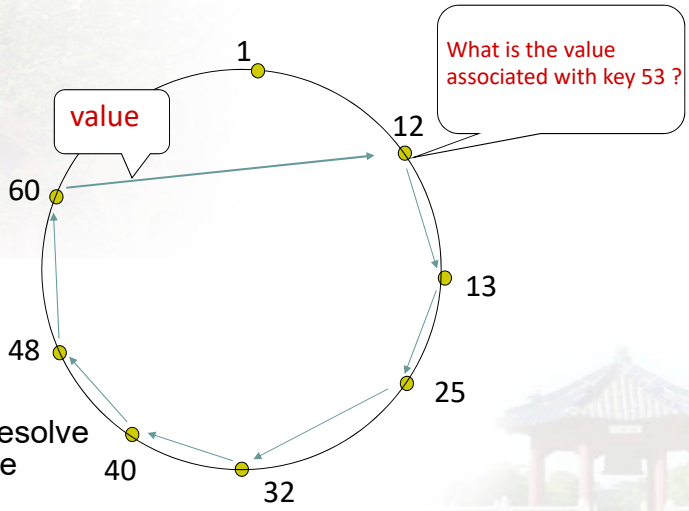
Resolving a query



O(N) messages on avgerage to resolve query, when there are *N* peers

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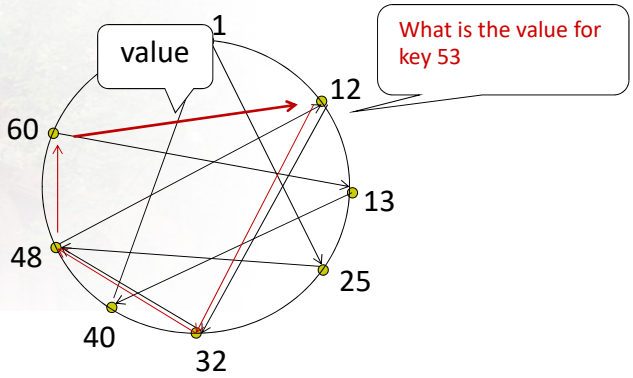
Circular DHT with shortcuts



- each peer keeps track of IP addresses of predecessor, successor, short cuts.
- reduced from 6 to 3 messages.
- possible to design shortcuts with *O(log N)* neighbors, *O(log N)* messages in query

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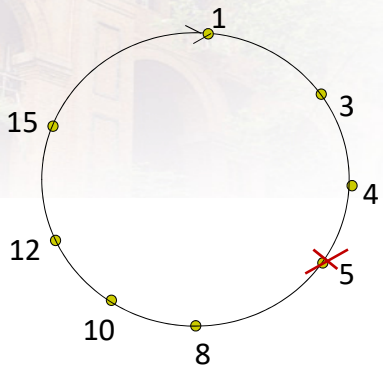
Peer churn



handling peer churn:

- peers may come and go (churn)
- each peer knows address of its two successors
- each peer periodically pings its two successors to check aliveness
- if immediate successor leaves, choose next successor as new immediate successor

example: peer 5 abruptly leaves

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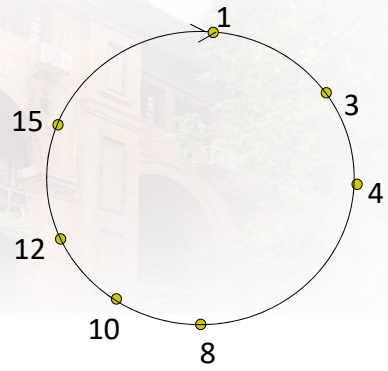
Peer churn



handling peer churn:

- peers may come and go (churn)
- each peer knows address of its two successors
- each peer periodically pings its two successors to check aliveness
- if immediate successor leaves, choose next successor as new immediate successor

example: peer 5 abruptly leaves

- **peer 4 detects peer 5's departure; makes 8 its immediate successor**
- **4 asks 8 who its immediate successor is; makes 8's immediate successor its second successor.**

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Chapter 2: outline

2.1 principles of network applications
- **app architectures**
- **app requirements**

2.2 Web and HTTP

2.3 FTP (cancel)

2.4 electronic mail
- **SMTP, POP3, IMAP**

2.5 DNS

2.6 P2P applications

2.7 video streaming and content distribution networks (CDNs)

2.8 socket programming with UDP and TCP

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Video Streaming and CDNs: context

- video traffic: major consumer of Internet bandwidth
  - Netflix, YouTube: 37%, 16% of downstream residential ISP traffic
  - ~1B YouTube users, ~75M Netflix users
- challenge: scale - how to reach ~1B users?
  - single mega-video server won't work (why?)
- challenge: heterogeneity
  - different users have different capabilities (e.g., wired versus mobile; bandwidth rich versus bandwidth poor)
- *solution:* distributed, application-level infrastructure

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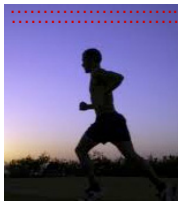
Multimedia: video

- **video: sequence of images displayed at constant rate**
  - **e.g., 24 images/sec**
- **digital image: array of pixels**
  - **each pixel represented by bits**
- **coding: use redundancy *within* and *between* images to decrease # bits used to encode image**
  - **spatial (within image)**
  - **temporal (from one image to next)**

spatial coding example: instead of sending *N* values of same color (all purple), send only two values: color value (*purple*) *and number of repeated values* (N)



frame *i*

frame *i+1*

temporal coding example: instead of sending complete frame at i+1, send only differences from frame i

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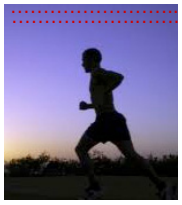
Multimedia: video

- CBR: (constant bit rate): video encoding rate fixed
- VBR: (variable bit rate): video encoding rate changes as amount of spatial, temporal coding changes
- examples:
  - MPEG 1 (CD-ROM) 1.5 Mbps
  - MPEG2 (DVD) 3-6 Mbps
  - MPEG4 (often used in Internet, < 1 Mbps)

spatial coding example: instead of sending *N* values of same color (all purple), send only two values: color value (*purple*) *and number of repeated values* (N)



frame *i*

frame *i+1*

temporal coding example: instead of sending complete frame at i+1, send only differences from frame i

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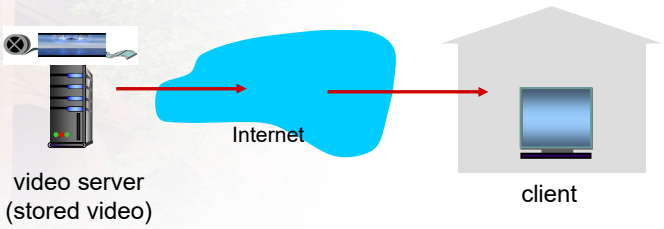
Streaming stored video:

simple scenario:



Main challenges:

- **server-to-client bandwidth will *vary* over time, with changing network congestion levels (in house, access network, network core, video server)**
- **packet loss, delay due to congestion will delay playout, or result in poor video quality**

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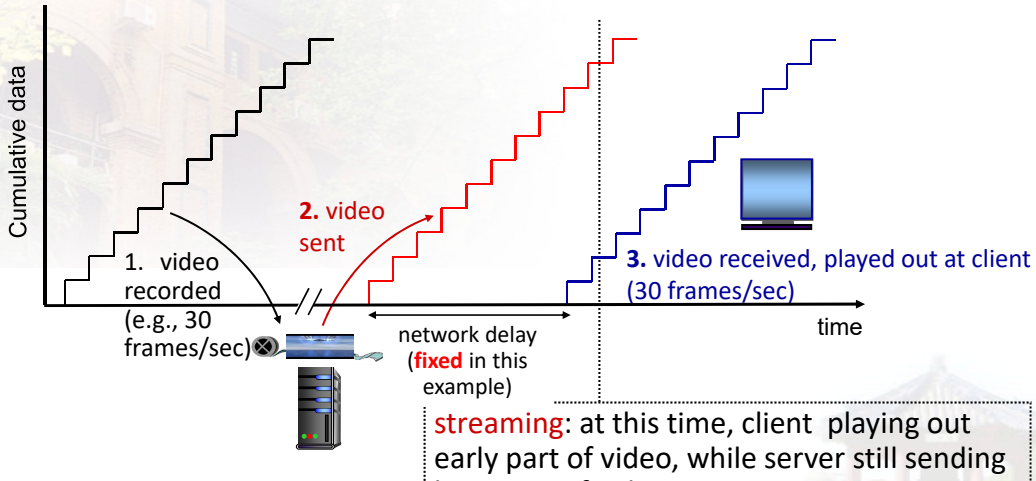
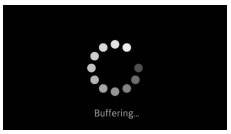
Streaming stored video



streaming: at this time, client playing out early part of video, while server still sending later part of video

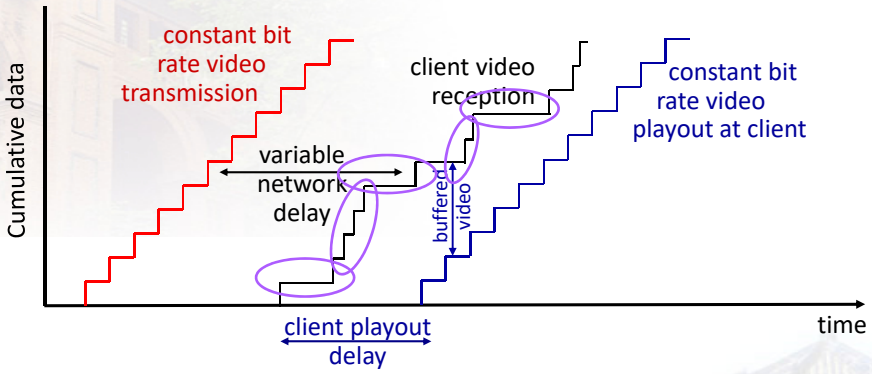
Streaming stored video: challenges

- continuous playout constraint: during client video playout, playout timing must match original timing
  - … but **network delays are variable** (jitter), so will need client-side buffer to match continuous playout constraint



- other challenges:
  - client interactivity: pause, fast-forward, rewind, jump through video
  - video packets may be lost, retransmitted

Streaming stored video: playout buffering



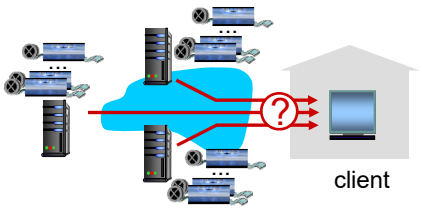
- *client-side buffering and playout delay:* compensate for network-added delay, delay jitter

Streaming multimedia: DASH

*D*ynamic, *A*daptive *S*treaming over *H*TTP

server:

- divides video file into multiple chunks
- each chunk encoded at multiple different rates
- different rate encodings stored in different files
- files replicated in various CDN nodes
- *manifest file:* provides URLs for different chunks

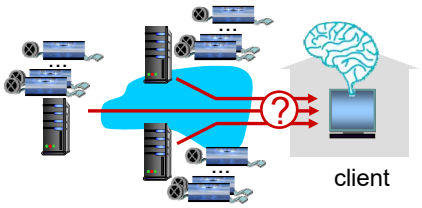


client:

- periodically estimates server-to-client bandwidth
- consulting manifest, requests one chunk at a time
  - chooses maximum coding rate sustainable given current bandwidth
  - can choose different coding rates at different points in time (depending on available bandwidth at time), and from different servers

Streaming multimedia: DASH

- *"intelligence"* at client: client determines
  - *when* to request chunk (so that buffer starvation, or overflow does not occur)
  - *what encoding rate* to request (higher quality when more bandwidth available)
  - *where* to request chunk (can request from URL server that is "close" to client or has high available bandwidth)



Streaming video = encoding + DASH + playout buffering

Content distribution networks

- ***challenge:*** **how to stream content (selected from millions of videos) to hundreds of thousands of *simultaneous* users?**

- ***option 1:*** **single, large "mega-server"**
  - **single point of failure**
  - **point of network congestion**
  - **long path to distant clients**
  - **multiple copies of video sent over outgoing link**

….quite simply: this solution *doesn't scale*

Content distribution networks

- ***challenge:*** **how to stream content (selected from millions of videos) to hundreds of thousands of simultaneous users?**
- ***option 2:*** **store/serve multiple copies of videos at multiple geographically distributed sites *(CDN)***
  - ***enter deep:*** **push CDN servers deep into many access networks**
    - **close to users**
    - **used by Akamai, 1700 locations**
  - ***bring home:*** **smaller number (10's) of larger clusters in POPs near (but not within) access networks**
    - **used by Limelight**

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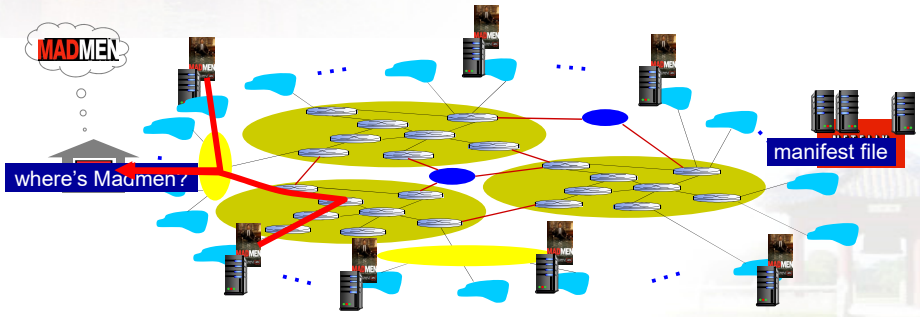
Content Distribution Networks (CDNs)

- CDN: stores copies of content at CDN nodes
  - e.g. Netflix stores copies of MadMen
- **subscriber requests content from CDN**
  - directed to nearby copy, retrieves content
  - may choose different copy if network path congested



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Content Distribution Networks (CDNs)

OTT challenges: **coping with a congested Internet**

- **from which CDN node to retrieve content?**
- **viewer behavior in presence of congestion?**
- **what content to place in which CDN node?**

more .. in chapter 7

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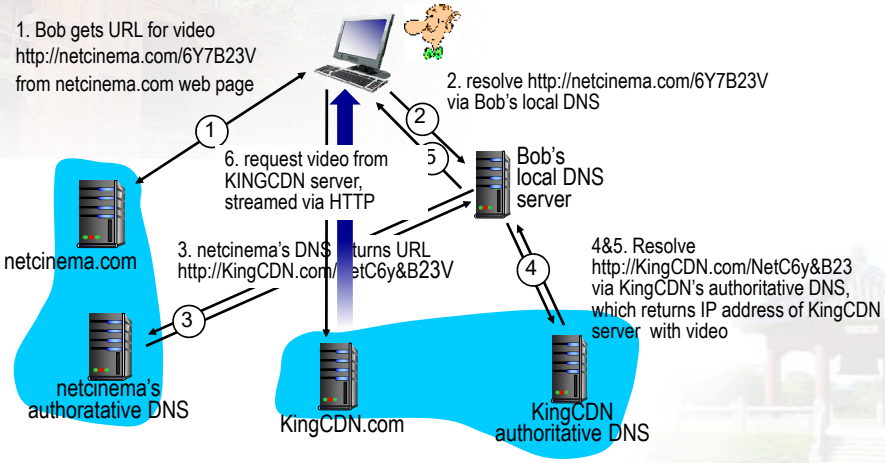
CDN content access: a closer look

Bob (client) requests video http://netcinema.com/6Y7B23V

- video stored in CDN at http://KingCDN.com/NetC6y&B23V



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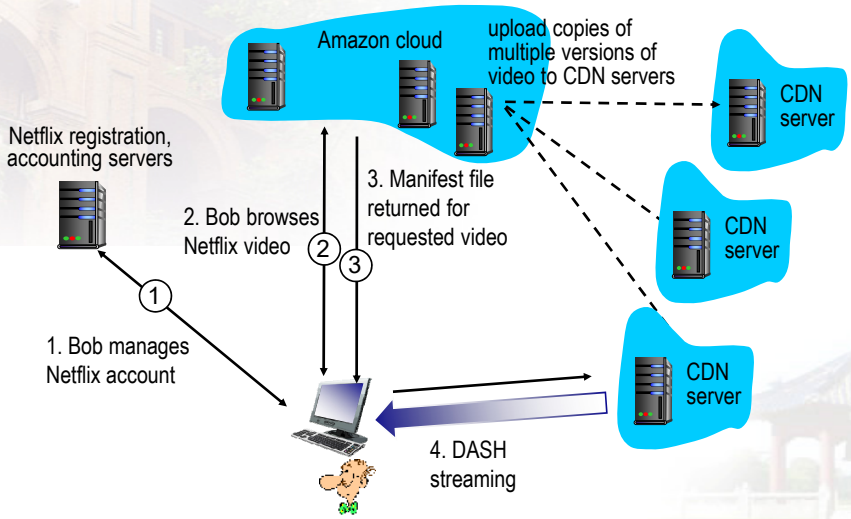
Case study: Netflix



Chapter 2: outline

2.1 principles of network applications
2.2 Web and HTTP
2.3 electronic mail
- SMTP, POP3, IMAP

2.4 DNS
2.5 P2P applications
2.6 video streaming and content distribution networks
2.7 socket programming with UDP and TCP (DIY)

Thanks

Q & A

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